



EÖTVÖS LORÁND UNIVERSITY

FACULTY OF INFORMATICS

DEPT. OF ALGORITHMS AND THEIR APPLICATIONS

Terraforming Mars in OpenGL

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Chapter 1

Introduction

During my studies, I learned object-oriented planning and software development, and have made it my own. I have come to truly enjoy thinking about and designing the architecture of software. As a result, I was keen on writing my thesis about a project where I could utilize and sharpen these skills.

Another subject I really enjoyed was the computer graphics course. What I liked most about it is how it required me to combine the knowledge I gained from all the other courses I took at the university. From linear algebra to algorithms to analysis and more, computer graphics uses parts from almost every computer science related field. I already knew after I finished the course, that I probably want to write my thesis about this topic.

To combine both of these into one, I chose to implement the board game Terraforming Mars in C++ with OpenGL. This way I had to design the model for the game logic, but also manage the view and graphics rendering of the application. The program would let the user play the solo version of Terraforming Mars with 3D graphics.

Chapter 2

User documentation

Chapter 3

Developer documentation

Chapter 4

Conclusion

Acknowledgements

Appendix A

Simulation results

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List of Algorithms

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